

Parity violation in β -decay of Cobalt-60

Parity or space inversion is a type of symmetry in quantum mechanics which converts a right-handed system (x, y, z) to a left-handed system $(-x, -y, -z)$.

π is defined to be the parity operator such that when a state is acted upon by π , it goes under a spatial inversion.

$$|\alpha\rangle \longrightarrow \pi|\alpha\rangle = |\alpha_p\rangle$$

Obviously, we require the expectation value of the position operator to gain a negative sign.

$$\langle\alpha_p|x|\alpha_p\rangle = \langle\alpha|\pi^\dagger x \pi|\alpha\rangle = -\langle\alpha|x|\alpha\rangle$$

$$\therefore \pi^\dagger x \pi = -x \quad \text{We define such an operator to be 'odd under parity'}$$

Further, evaluating sequentially from the inside out:

$$\pi^\dagger(\pi^\dagger x \pi)\pi = \pi^\dagger(-x)\pi = -(\pi^\dagger x \pi) = -(-x) = x \implies \pi^2 = \mathbb{I} \text{ and } (\pi^\dagger)^2 = \mathbb{I}$$

From $\pi^2 = \mathbb{I}$, we have $\pi = \pi^{-1}$

From π being unitary, we have

$$\pi = \pi^{-1} = \pi^\dagger$$

Hence, π is Hermitian and unitary.

The fundamental commutation relation should not break under π :

$$\begin{aligned} [x_i, p_j] &= i\hbar\delta_{ij} \\ \pi[x_i, p_j]\pi^\dagger &= i\hbar\delta_{ij}\pi\pi^\dagger \\ [-x_i, \pi p_j \pi^\dagger] &= i\hbar\delta_{ij} \end{aligned}$$

Hence, we arrive at $\pi p \pi^\dagger = -p$ to keep the commutation relation preserved. Those vectors which are odd under parity are called polar vectors.

Now coming to orbital angular momentum,

$$L = x \times p = i\hbar\epsilon_{ijk}x_j\partial_k = \epsilon_{ijk}x_jp_k$$

Since it is constructed of two odd operators,

$$\pi L \pi^\dagger = L \quad : \text{ defined as even under parity}$$

Further, using the fact that the generator of rotations and the matrix representation of parity commute, we state:

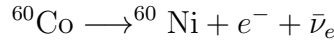
$$\begin{aligned} [\pi, J] &= 0 \\ \pi J &= J\pi \\ \therefore J &= \pi^\dagger J \pi \quad (\text{called pseudovector}) \end{aligned}$$

Since intrinsic spin S obeys the exact same commutation relations as J (e.g., $[S_i, S_j] = i\hbar\epsilon_{ijk}S_k$), it transforms identically to J as a pseudovector under parity.

Hence, J , L , and S are even under parity and by convention are called pseudovectors.

We have now defined sufficient background to understand the origin of parity non-conservation proposed by Lee and Yang and confirmed by the experiment of C.S. Wu using supercold, polarized Cobalt-60 isotope.

The Cobalt-60 isotope is expected to show β -decay due to it being unstable. It is cooled down to minimize any emission due to thermal fluctuations and also polarized by an external magnetic field such that the angular momentum vector J points in a defined direction.



The external magnetic field (homogeneous and uniform) is to rotate the nuclear spin for it to align with the direction of the magnetic field and also to attain uniform Zeeman splitting.

The decaying e^- are shot out approximately in the north direction (along the direction of the spin angular momentum vector). Under space inversion (more loosely referred to as looking into the mirror world), both the momentum of e^- and S behave differently.

NOTE: Mirror world analogy generally corresponds to only the action of reflection whereas inversion is reflection plus a 180° turn.

Under parity transformation:

$$\pi^\dagger p \pi = -p \quad \pi^\dagger S \pi = S$$

Hence, in the inverted world, the shot out e^- are now in the opposite direction with respect to the spin angular momentum vector. This was the groundbreaking asymmetry found in weak interactions.

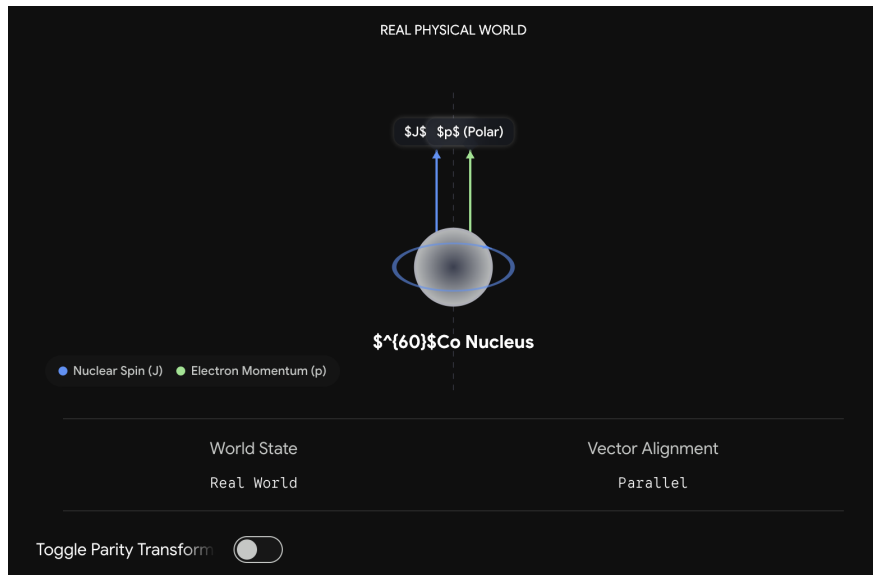


Figure 1: Ordinary World. Momentum vector and Spin angular momentum vector are parallel

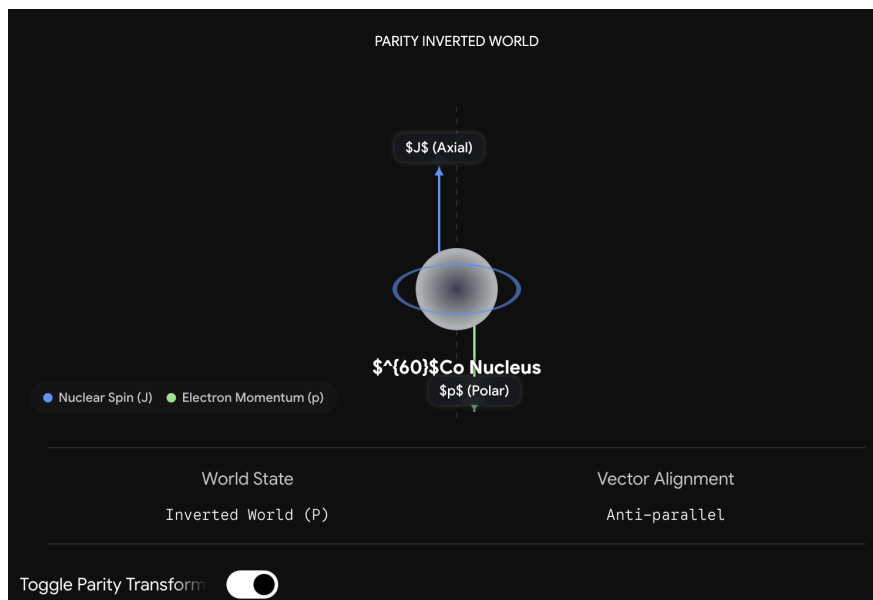


Figure 2: Inverted World. Momentum vector and Spin angular momentum vector are anti-parallel